Two spins coupled to an XXZ bulk

Reduced channels and quantum-state texture

Spin transport project

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Two-probe reduced dynamics

Two spin-1/2 probes, A and B, attach to opposite ends of a finite XXZ bulk. The total Hamiltonian drives the dynamics.

- Ising contacts give an exact time-dependent Schur channel.
- Exchange contacts admit a magnetization-based block reduction under explicit symmetry assumptions.
- The texture study uses only $|+\rangle_A \otimes |+\rangle_B$ as probe input.

The bulk preparation determines the time-dependent channel coefficients.

Hamiltonian and normalization

N bulk spins, $N+2$ total. Physical order: $A, 1, \ldots, N, B$.

$$S^\alpha = \frac{\sigma^\alpha}{2}, \quad S^\pm = S^x \pm i S^y, \quad \hbar = 1.$$

$$H_E = J \sum_{j=1}^{N-1} \left(S_j^x S_{j+1}^x + S_j^y S_{j+1}^y + \Delta S_j^z S_{j+1}^z \right).$$

$$\begin{aligned} V_{\mathrm{ZZ}} &= g_A S_A^z S_1^z + g_B S_N^z S_B^z, \\ V_{\mathrm{ex}} &= \frac{g_A}{2}(S_A^+ S_1^- + S_A^- S_1^+) + \frac{g_B}{2}(S_N^+ S_B^- + S_N^- S_B^+), \\ H^{(r)} &= H_E + V_r, \qquad r \in \{\mathrm{ZZ}, \mathrm{ex}\}. \end{aligned}$$

J, g_A, g_B have energy units. These are alternative contact models, with no direct A–B interaction and no applied fields in the reference model.

A fixed product preparation defines a channel

Group the Hilbert space as $\mathcal{H}_{AB} \otimes \mathcal{H}_E$. Fix the same bulk density matrix ω_E for every probe input.

$$\rho(0) = \rho_{AB}(0) \otimes \omega_E, \qquad U_t = e^{-iH^{(r)}t}.$$

$$\boxed{\Lambda_t(X) = \mathrm{Tr}_E[U_t(X \otimes \omega_E)U_t^\dagger].}$$

This map is completely positive and trace preserving (CPTP).

- The channel acts on the joint two-probe input, of dimension four.
- Changing ω_E changes the channel.
- The derivation uses exact unitary dynamics and a partial trace.

Ising coupling: four conditional bulk Hamiltonians

Both probe projections are conserved:

$$[H^{(\mathrm{ZZ})}, S_A^z] = [H^{(\mathrm{ZZ})}, S_B^z] = 0.$$

Label the probe basis by $a = (s_A, s_B)$, $s_A, s_B = \pm\frac{1}{2}$, and set $P_a = |a\rangle\langle a|$.

$$\boxed{H^{(\mathrm{ZZ})} = \sum_a P_a \otimes H_a, \qquad H_a = H_E + g_A s_A S_1^z + g_B s_B S_N^z.}$$

Since $P_a P_b = \delta_{ab} P_a$,

$$U_t = \sum_a P_a \otimes U_a(t), \qquad U_a(t) = e^{-iH_a t}.$$

The input can coherently occupy all four branches.

Ising coupling: derivation of the Schur channel

For each matrix unit $E_{ab} = |a\rangle\langle b|$,

$$P_c E_{ab} P_d = \delta_{ca}\delta_{db} E_{ab}.$$

Only one conditional pair survives the partial trace:

$$\Lambda_t(E_{ab}) = \Gamma_{ab}(t) E_{ab}, \qquad \Gamma_{ab}(t) = \mathrm{Tr}_E[U_a(t)\omega_E U_b(t)^\dagger].$$

By linearity,

$$\boxed{\Lambda_t(X) = \Gamma(t) \circ X.}$$

For a pure bulk, $\Gamma_{ab} = \langle\phi_0|U_b^\dagger U_a|\phi_0\rangle$. Its complex phase is part of the channel data.

Ising channel constraints

$$\Gamma = \Gamma^\dagger \succeq 0, \qquad \Gamma_{aa} = 1, \qquad |\Gamma_{ab}| \leq 1.$$

These follow from a Gram factorization of the conditional bulk amplitudes.

- All four computational-basis populations stay fixed.
- The map is unital: $\Lambda_t(I_4) = I_4$.
- Six complex coherence factors specify its nontrivial action.
- A common bulk can correlate the two probes.

This is generalized **dephasing**. A depolarizing map $\lambda X + (1-\lambda)\,\mathrm{Tr}(X) I_4/4$ fixes every population only at $\lambda = 1$, the identity.

The $|{+}{+}\rangle$ output determines the Ising channel

$$|f_1\rangle = |{+}{+}\rangle = \frac{1}{2}\sum_{a=1}^{4}|a\rangle, \qquad f_1 = |f_1\rangle\langle f_1|, \qquad (f_1)_{ab} = \frac{1}{4}.$$

The full output is $\rho_f(t) = \Lambda_t(f_1)$, so

$$\boxed{\rho_f(t) = \frac{\Gamma(t)}{4}, \qquad \Gamma(t) = 4\rho_f(t).}$$

$$\Lambda_t(X) = [4\rho_f(t)] \circ X.$$

One known preparation and *full output tomography* identify the instantaneous channel. The scalar texture is only part of this information.

Ising channel: normalized Choi state and Kraus operators

Use output–reference order and $|\Phi_4\rangle = \frac{1}{2}\sum_a |aa\rangle$:

$$J_t = (\Lambda_t \otimes \mathrm{id})(|\Phi_4\rangle\langle\Phi_4|) = \frac{1}{4}\sum_{a,b} \Gamma_{ab}|aa\rangle\langle bb|.$$

With $W|a\rangle = |aa\rangle$,

$$\boxed{J_t = W\rho_f(t)W^\dagger, \qquad \operatorname{rank} J_t = \operatorname{rank}\Gamma \le 4.}$$

The Choi eigenvalues are those of ρ_f, with twelve extra zeros.

$$S(J_t) = S(\rho_f), \qquad \mathrm{Tr}\, J_t^2 = \mathrm{Tr}\,\rho_f^2 = \frac{1}{16}\sum_{a,b} |\Gamma_{ab}|^2.$$

If $\Gamma = \sum_k \lambda_k v_k v_k^\dagger$, a minimal Kraus family is $D_k = \sqrt{\lambda_k}\,\mathrm{diag}(v_k)$.

Ising channel: unitarity and loss of reference entanglement

The map is unitary iff $\operatorname{rank}\Gamma = 1$:

$$\Gamma_{ab} = u_a u_b^*, \quad |u_a| = 1, \qquad \Lambda_t(X) = \operatorname{diag}(u) X \operatorname{diag}(u)^\dagger.$$

The partial-transpose eigenvalues of its Choi state are

$$\frac{1}{4} \text{ (four times)}, \qquad \pm\frac{|\Gamma_{ab}|}{4} \quad (a < b).$$

Therefore it is entanglement breaking iff

$$\Gamma_{ab} = 0 \quad \text{for every } a \neq b.$$

This is entanglement between the probe pair and a reference. Entanglement of A with B is a separate question.

Ising channel: exact tests between different times

The superoperator eigenvalues are $\Gamma_{ab}(t)$. It is invertible iff every factor is nonzero.

$$\mathcal{L}_t(E_{ab}) = \frac{\dot{\Gamma}_{ab}(t)}{\Gamma_{ab}(t)} E_{ab} \qquad \text{at regular times.}$$

If Λ_s is invertible, define

$$Q_{ab}(t,s) = \frac{\Gamma_{ab}(t)}{\Gamma_{ab}(s)}.$$

$$\boxed{\Lambda_t \Lambda_s^{-1} \text{ is CPTP} \iff Q(t,s) \succeq 0.}$$

At singular times this division is undefined. A coherence factor that vanishes and later revives prevents divisibility through that time.

Ising channel: maximum concurrence from separable inputs

For the chosen fixed bulk preparation,

$$\boxed{\max_{\rho \in \mathrm{Sep}(A:B)} C[\Lambda_t(\rho)] = C[\rho_f(t)] = C[\Gamma(t)/4].}$$

Proof idea. A pure product input with amplitudes $(a_0, a_1) \otimes (b_0, b_1)$ gives a local diagonal filter of ρ_f. The concurrence transforms as

$$C_{\mathrm{out}} = 4|a_0 a_1 b_0 b_1| C(\rho_f) \leq C(\rho_f).$$

Convexity extends the bound to separable mixtures. The preparation $|{+}{+}\rangle$ attains it. Thus ρ_f is entangled iff this channel can entangle some separable input.

Exchange coupling: total magnetization sectors

Define

$$M_S = S_A^z + S_B^z, \quad M_E = \sum_{j=1}^{N} S_j^z, \quad M = M_S + M_E.$$

For the field-free model,

$$[H^{(\mathrm{ex})}, M] = 0, \qquad \dim \mathcal{H}_m = \binom{N+2}{(N+2)/2 - m}.$$

- Exchange transfers spin between a probe and the bulk.
- Probe populations and coherences can both change.
- The four conditional-Hamiltonian construction does not apply.

These Hamiltonian sectors exist for any input. A reduced-channel symmetry also needs a condition on the bulk preparation.

Exchange channel: the bulk assumption and covariance

Assume the fixed bulk preparation satisfies

$$\boxed{[\omega_E, M_E] = 0.}$$

A single bulk magnetization sector, or an incoherent mixture of sectors, satisfies this condition.

$$V_\theta = e^{-i\theta M_S}, \quad W_\theta = e^{-i\theta M_E}, \quad [U_t, V_\theta \otimes W_\theta] = 0.$$

Moving these rotations through the propagator and partial trace gives

$$\boxed{\Lambda_t(V_\theta X V_\theta^\dagger) = V_\theta \Lambda_t(X) V_\theta^\dagger.}$$

This covariance preserves coherence order. It permits a change in probe magnetization and does not require a unital channel.

Exchange channel: five coherence-order blocks

The basis charges are $(m_{00}, m_{01}, m_{10}, m_{11}) = (1, 0, 0, -1)$. An operator E_{ab} has order $q = m_a - m_b$.

$$\mathcal{V}_q = \mathrm{span}\{E_{ab} : m_a - m_b = q\}, \qquad \Lambda_t(\mathcal{V}_q) \subseteq \mathcal{V}_q.$$

q	Dimension	Content
0	6	Four populations and $01 \leftrightarrow 10$ coherences
$+1, -1$	4 each	Single-unit coherences
$+2, -2$	1 each	$00 \leftrightarrow 11$ coherences

$$\boxed{\Lambda_t \simeq T^{(-2)} \oplus T^{(-1)} \oplus T^{(0)} \oplus T^{(+1)} \oplus T^{(+2)}.}$$

The block dimensions are $1, 4, 6, 4, 1$. Their entries remain dynamical quantities.

Exchange channel: Choi blocks and charge-transfer Kraus operators

For the normalized Choi state in output–reference order,

$$[J_t, M_S \otimes I - I \otimes M_S^{\mathrm{T}}] = 0.$$

Grouping $|c\rangle_{\mathrm{out}}|a\rangle_R$ by $m_c - m_a$ also gives Choi block sizes $1, 4, 6, 4, 1$. Choose bulk eigenstates of magnetization in the spectral decomposition $\omega_E = \sum_\nu p_\nu |\phi_\nu\rangle\langle\phi_\nu|$ and output basis $|r\rangle$. Then

$$K_{r\nu} = \sqrt{p_\nu}\, {}_E\langle r|U_t|\phi_\nu\rangle_E, \qquad [M_S, K_{r\nu}] = (\mu_\nu - \mu_r) K_{r\nu}.$$

The two probes permit charge transfers only in $\{-2, -1, 0, 1, 2\}$. Choi blocks and superoperator blocks are different rearrangements of the coefficients. Equal dimensions do not give equal spectra.

A fixed total charge gives a restricted state trajectory

If the *global* preparation lies in one total-M sector, orthogonality of the conditional bulk charges gives

$$\rho_{AB}(t) = \begin{pmatrix} p_{00} & 0 & 0 & 0 \\ 0 & p_{01} & c & 0 \\ 0 & c^* & p_{10} & 0 \\ 0 & 0 & 0 & p_{11} \end{pmatrix}.$$

$$C = 2\max\{0, |c| - \sqrt{p_{00}p_{11}}\}.$$

For fixed bulk charge, $|{+}{+}\rangle$ instead occupies *three* total-charge sectors. Its reduced output need not have this restricted X form.

Quantum-state texture and the return probability

In the computational basis, prepare the probes in $f_1 = |{+}{+}\rangle\langle{+}{+}|$. For its output $\rho_f(t) = \Lambda_t(f_1)$, define

$$\Sigma(\rho_f) = \sum_{a,b} (\rho_f)_{ab}, \qquad p_f = \langle{+}{+}|\rho_f|{+}{+}\rangle = \frac{\Sigma(\rho_f)}{4}.$$

$$\boxed{\Omega_{T_F} = 1 - p_f, \qquad \Omega_R = -\ln p_f.}$$

Initially $p_f = 1$. Always $0 \le p_f \le 1$. Rugosity diverges at $p_f = 0$. These definitions apply in both coupling regimes. Texture depends on phases in the chosen basis and is distinct from purity and probe entanglement.

One measurement setting gives the grand sum

Since $4f_1 = (I + \sigma_A^x)(I + \sigma_B^x)$,

$$\boxed{\Sigma(\rho_f) = 1 + \langle \sigma_A^x \rangle + \langle \sigma_B^x \rangle + \langle \sigma_A^x \sigma_B^x \rangle.}$$

Measure both probes along x:

$$p_f = P_{++}^{(x)}.$$

- This setting gives the texture scalar in either regime.
- The Ising channel requires six complex output coherences for full reconstruction. Eight local Pauli settings suffice.
- All measurements use the same input preparation $|{+}{+}\rangle$.

Ising texture gives two channel fidelities

Using $\rho_f = \Gamma/4$,

$$p_f = \frac{1}{16}\sum_{a,b}\Gamma_{ab} = \frac{1}{4} + \frac{1}{8}\sum_{a<b}\Re\Gamma_{ab}.$$

The entanglement fidelity relative to the identity satisfies

$$F_e = \langle\Phi_4|J_t|\Phi_4\rangle = p_f.$$

Consequently,

$$\boxed{F_{\text{avg}} = \frac{4F_e + 1}{5} = \frac{1 + \Sigma(\rho_f)}{5}.}$$

These concern the two-probe channel relative to the identity. They are not an A-to-B state-transfer fidelity.

Ising texture: response and preservation

For any channel, its grand-sum response operator is

$$\mathcal{R}_t = 4\Lambda_t^*(f_1), \qquad \Sigma[\Lambda_t(X)] = \mathrm{Tr}(\mathcal{R}_t X).$$

For the Ising Schur map,

$$\boxed{\mathcal{R}_t = \Gamma(t)^{\mathsf{T}}.}$$

The transpose is ordinary. Within this family, either $\Lambda_t(f_1) = f_1$ or $\Lambda_t^*(f_1) = f_1$ forces every $\Gamma_{ab} = 1$. Thus vanishing fidelity texture identifies the identity channel within the known Ising family.

Exchange texture: what the same input determines

With a magnetization-invariant bulk, f_1 probes all five coherence orders. One output gives the action of each block on *one vector* in that block. The 6- and 4-dimensional blocks remain undetermined. The one-dimensional highest-order sector is an exception:

$$\Lambda_t(E_{00,11}) = \eta_{+2}(t) E_{00,11}, \qquad \boxed{\eta_{+2}(t) = 4[\rho_f(t)]_{00,11}.}$$

The scalar texture remains $1 - p_f$ or $-\ln p_f$. Its equality with a channel identity-fidelity expression requires the Ising Schur structure and is not a general exchange result.

Applicability of the symmetry reduction

The exchange block decomposition needs both

$$[H, M_S + M_E] = 0 \qquad \text{and} \qquad [\omega_E, M_E] = 0.$$

- Transverse bulk fields generally violate total-magnetization conservation.
- Coherence between bulk magnetization sectors generally removes channel phase covariance, even for a conserving Hamiltonian.
- The Ising Schur channel and $\Gamma = 4\rho_f$ remain valid in both cases, because the probe z components are still conserved.

With transverse bulk fields, exchange dynamics generally couples different total-magnetization sectors.

Channel structure and dynamical coefficients

	Ising contacts	Exchange contacts
Map structure	$\Gamma \circ X$	Five coherence-order blocks under the symmetry assumptions
Populations	Fixed	Allowed
Information in ρ_f	Complete channel	Partial block information
Coefficients	Six complex conditional overlaps	Entries of the symmetry blocks

The symmetry reduction fixes the allowed mixing of probe operators. Their time-dependent coefficients require the bulk dynamics.

Entanglement conventions and references

Probe entropy is $S(\rho) = -\operatorname{Tr}\rho \ln \rho$ in nats. For a globally pure state, $S(\rho_{AB})$ measures entanglement across $AB|E$. Probe concurrence and negativity concern $A|B$. The normalized Choi state has trace one. Its reference–output entanglement is distinct from entanglement between the two probes.

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